

BESS OPERATIONS REPORT

April 2026

Site:	Nur Bukhara 63MW/126MWh BESS Plant
Reporting period:	01 April 2026 — 30 April 2026
Blocks monitored:	15 blocks
Report generated:	2026-05-17 09:38
Report type:	Monthly Operations & Performance Summary

Report generated automatically from SCADA exports. Per-block KPIs use the validated availability engine. RTE / EFC / throughput sourced from cumulative energy counters.

1. Project Details

Parameter	Value
Project name	Nur Bukhara 63MW/126MWh BESS Plant
Reporting period	01 April 2026 — 30 April 2026
OEM	Sungrow Power Supply Co. Ltd.
Number of blocks	15 blocks
Customer	MASDAR
O&M Company	MASDAR MSTs
OEM	Sungrow Power Supply Co. Ltd.
Project Capacity	63 MW / 126 MWh
Equipment	ST5015UX-S-2H_V152 (30); SC5500UD-MV_V129 (15); SCC_V152 (15)

2. Summary

During the reporting period **01 April 2026 — 30 April 2026**, the Nur Bukhara 63MW/126MWh BESS Plant fleet (15 blocks) delivered **6,929.1 MWh** of discharge energy on **7,509.5 MWh** of charge energy. Site round-trip efficiency was **92.27%** (above the 85% target). Plant-level availability (redundancy-based) was **52.08%**; container-level availability was **77.91%**. Average state of charge: **48.28%**. Average state of health: **97.86%**. **705** equivalent full cycles were completed across the fleet (avg 47.0 per block, 7.8% of the 600/year contractual target). **1.9** day(s) per block (fleet average) were flagged as EXCLUDED and removed from RTE / availability calculations.

3. Services Provision

3.1 Preventative Maintenance (PM)

No PM activities in the reporting period.

3.2 Corrective Maintenance

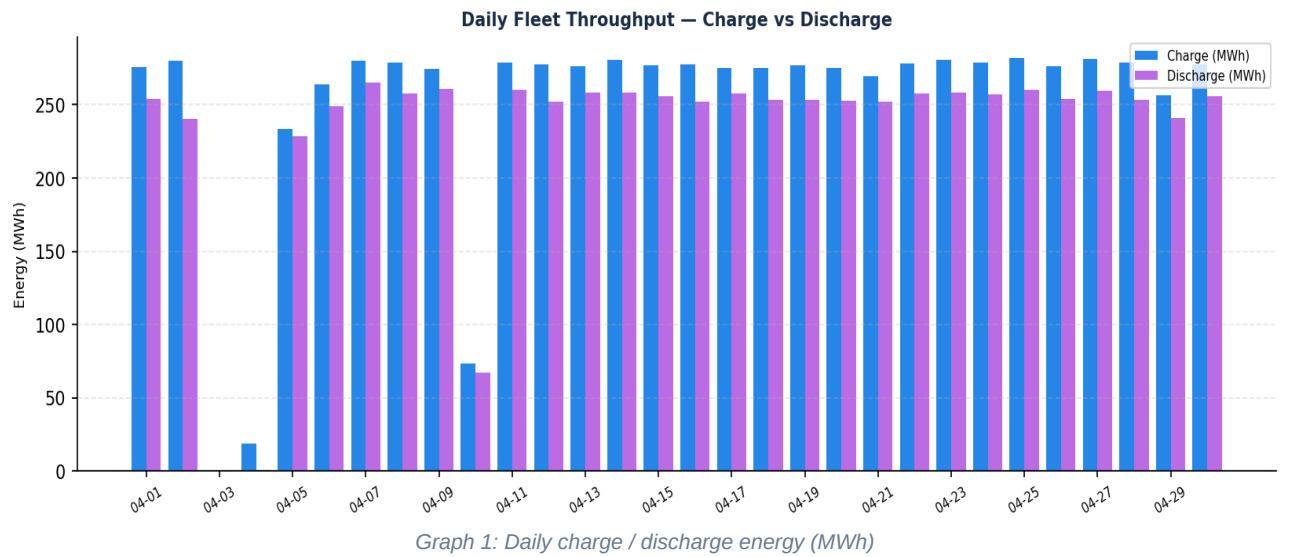
Automated summary based on alarm classification — recurring corrective items observed this period:

Item	Occurrences	Resolution applied
Manual Stop	1356	No action required (manual operator command)
BSC Unit Fault	667	Restart BSC; check BMS comm
LC System Fault	579	Restart LC; check downstream comm
LC-PCS Communication	402	Check ethernet between LC and PCS
FSS Fault	383	Fire Safety System check required

4. Plant's Performance

4.1 Plant Performance Data

No	Item	Total (MWh)
1	Imported Energy (Charge)	7,509.47
2	Exported Energy (Discharge)	6,929.11



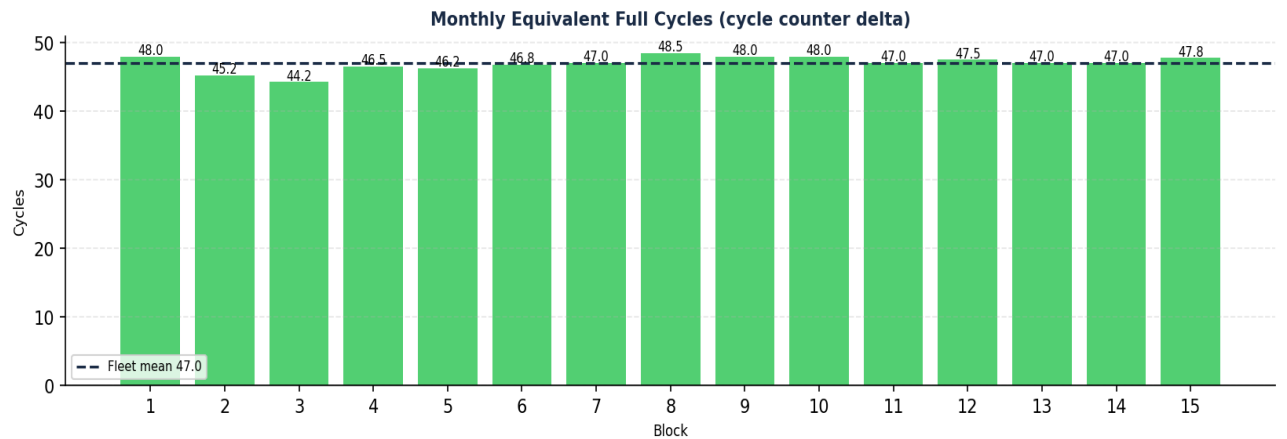
Number of Cycles

No	Item	Total	% of yearly target (600)
1	Number of Cycles in Reported Month	47.0	7.83%
2	Accumulative Number of Cycles in one Year	301.0	50.17%

Cycles Block-wise

Block	Accumulative cycles	Cycles in period	Charged Energy (MWh)	Discharged Energy (MWh)
Block 1	294.00	48.00	510.2	470.2
Block 2	291.25	45.25	499.9	460.6
Block 3	290.25	44.25	473.5	436.1
Block 4	289.25	46.50	495.8	457.2
Block 5	290.50	46.25	493.3	454.9
Block 6	293.25	46.75	496.3	457.8
Block 7	292.25	47.00	497.5	459.0
Block 8	292.50	48.50	509.9	470.2
Block 9	290.75	48.00	507.2	467.6
Block 10	287.75	48.00	523.3	487.9
Block 11	282.75	47.00	483.9	446.1
Block 12	292.50	47.50	502.0	463.0

Block	Accumulative cycles	Cycles in period	Charged Energy (MWh)	Discharged Energy (MWh)
Block 13	289.00	47.00	502.2	463.1
Block 14	293.50	47.00	508.0	468.5
Block 15	291.25	47.75	506.5	467.1



Graph 2: Monthly EFC per block (cycle counter delta)

Availability Hours (plant-level redundancy method)

Item	Value
Scheduled hours	719.5 h
Scheduled unavailability (PM)	0.0 h
Unscheduled outage (container-aggregate)	2,688.9 h (373.72%)
Plant-level outage (redundancy adjusted)	344.8 h (47.92%)
Plant-level availability	52.08%
Container-level availability (current engine)	77.91%

4.2 Key Performance Indicators (KPIs)

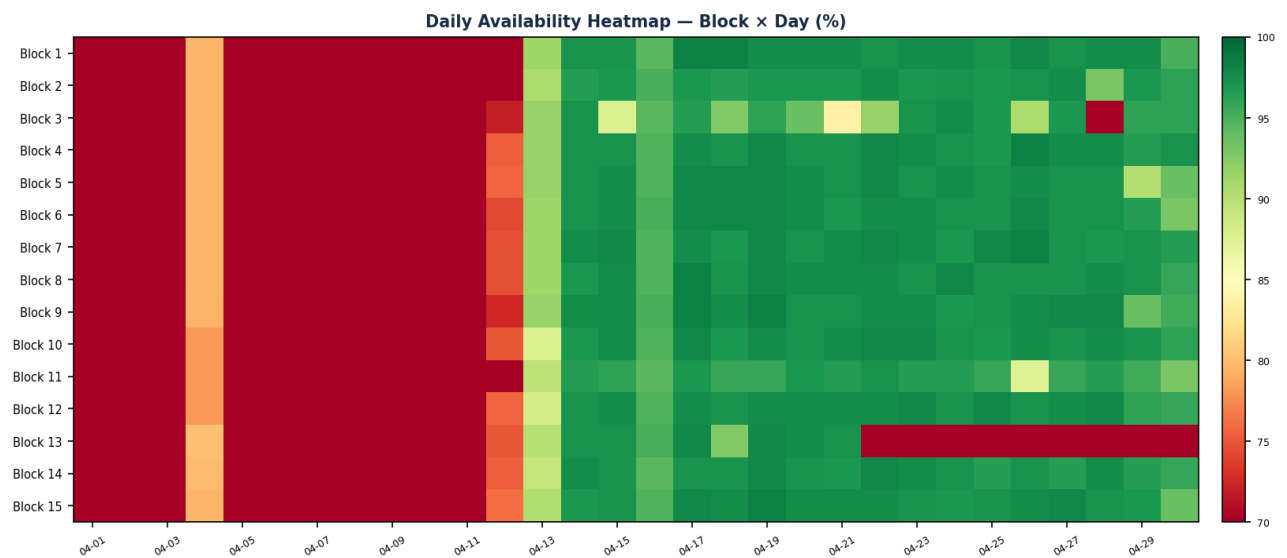
48.28%	97.86%	92.27%
Average SOC	Average SOH	RTE

4.3 Monthly Performance Comparison

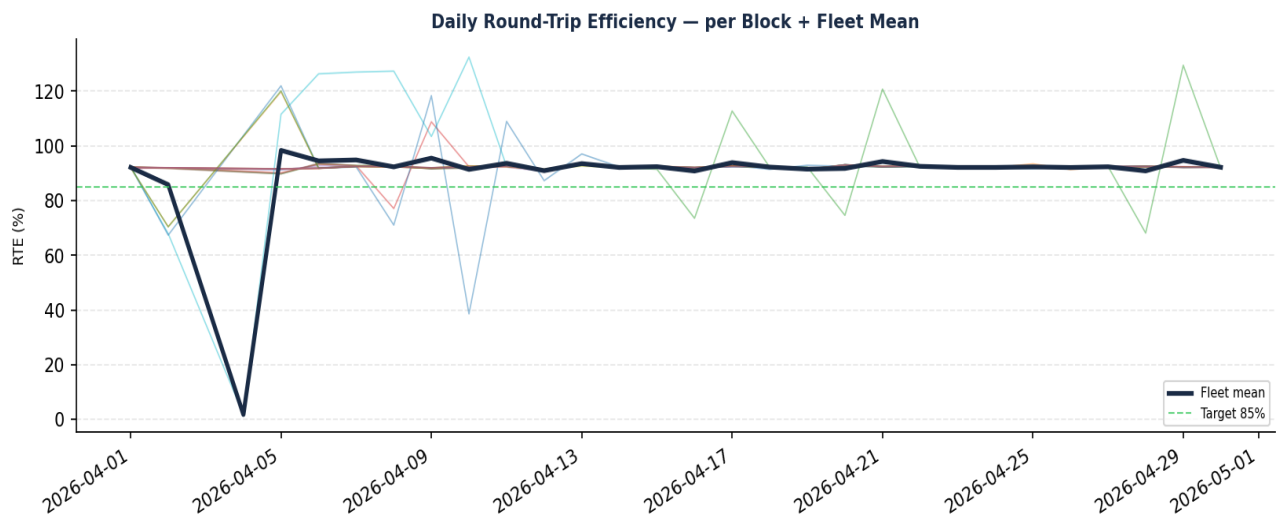
Month	SOC	SOH	RTE	Cycles	Discharge (MWh)	Charge (MWh)
December 2025	41.0%	99.0%	90.0%	—	2,843	3,146
January 2026	56.0%	99.0%	90.0%	—	6,441	7,122
February 2026	48.0%	99.0%	90.0%	—	5,728	6,336

Month	SOC	SOH	RTE	Cycles	Discharge (MWh)	Charge (MWh)
April 2026	48.3%	97.9%	92.3%	—	6,929	7,509
March 2026	47.1%	99.2%	93.0%	—	30,868	33,179
April 2026	48.3%	97.9%	92.3%	47.0	6,929	7,509

4.4 System Availability



Graph 3: Daily availability heatmap — block x day. Red = outage, green = full operation. Quality flags applied: EXCLUDED days (zero throughput) are removed from the calculation.



Graph 4: Daily RTE per block (light lines) and fleet mean (dark line). Fleet mean target: 85%.

Block performance anomalies (3 block(s) flagged as more than 1σ below fleet mean on RTE or throughput):

Block	Avg RTE	RTE z-score	Avg Throughput	Avg Availability	Total Alarms
Block 3	89.68%	-2.57	33.0 MWh	72.86%	206
Block 11	90.33%	-1.74	34.5 MWh	80.12%	129
Block 5	92.18%	+0.61	33.9 MWh	77.70%	212

5. System Operation

5.1 Summary of Faults and Alarms

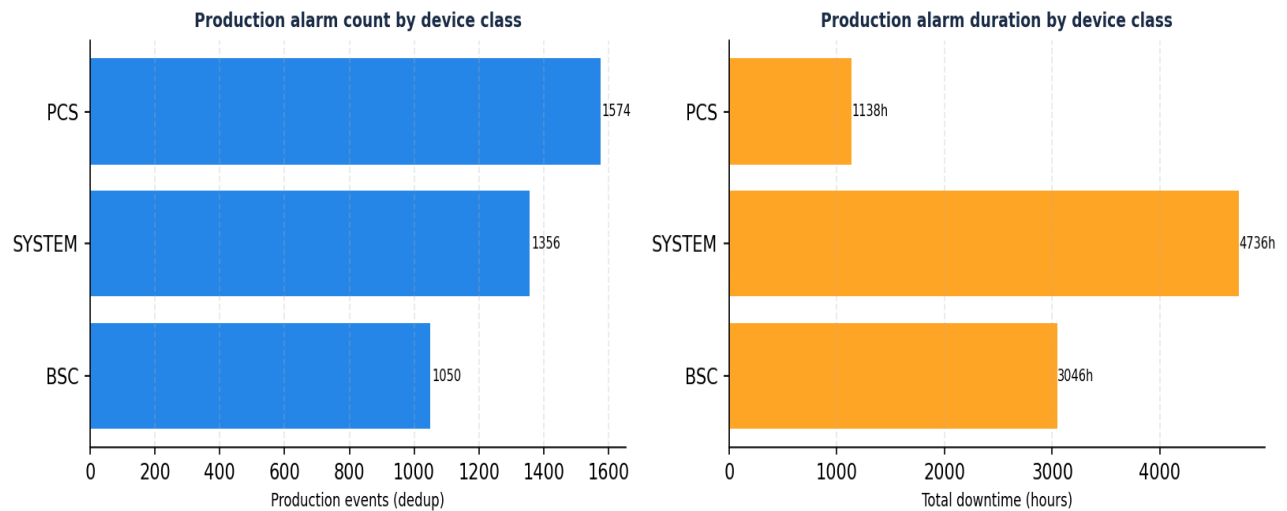
Faults (Production)

Sub-system	Reason / Cause	Occurrences	Total Hours	Blocks Affected	Resolution
PCS	Manual Stop	1356	1098.9 h	01, 02, 03, 04, 05, 06, 07, 08	No action required (manual operator comm)
BESS	BSC Unit Fault	667	2454.1 h	01, 02, 03, 04, 05, 06, 07, 08	Restart BSC; check BMS comm
System	LC System Fault	579	1835.2 h	01, 02, 03, 04, 05, 06, 07, 08	Restart LC; check downstream comm
SCC+PCS	LC-PCS Communication	402	1601.7 h	01, 02, 03, 04, 05, 06, 07, 08	Check ethernet between LC and PCS
BESS	FSS Fault	383	591.6 h	01, 02, 03, 04, 05, 06, 07, 08	Fire Safety System check required
SCC+BESS	LC-BSC Communication	369	1297.4 h	01, 02, 03, 04, 05, 06, 07, 08	Check ethernet between LC and BSC
PCS	AC Switch Fault	112	9.3 h	05, 07, 11	Reboot PCS from web-HMI
PCS	Soft Start (DC switch) Fault	75	6.8 h	03, 04, 05, 06, 09, 11, 12, 13	Self-reset on next charge/discharge
PCS	Island Protection	23	22.2 h	01, 02, 03, 04, 05, 07, 09, 11	Grid sync check required
PCS	PCS Unit Fault	4	0.9 h	01	Investigate PCS logs
PCS	AC Over-voltage	3	0.2 h	05, 06, 11	Grid voltage check
PCS	PCS LCD-DSP Comm	3	0.2 h	14	Restart PCS HMI
PCS	PCS Fault (general)	1	0.6 h	01	Investigate PCS logs
PCS	PCS Cooling Fan	1	0.1 h	09	Replace fan / check airflow
PCS	PCS SCU-DSP Comm	1	0.4 h	05	Restart PCS controller

Alarms (Persistent Warnings)

Sub-system	Reason / Cause	Occurrences	Total Hours
PCS	Inverter Fault	3309	218.9 h
BESS	Battery Fault	1744	5537.2 h
BESS	LCU Communication	795	3168.7 h
BESS	Humiture Sensor Comm	794	3167.2 h
BESS	Auxiliary Sub-Device	793	3467.5 h
BESS	Dehumidifier Comm	740	2950.6 h
BESS	Gas Detector Comm	611	2493.1 h

Sub-system	Reason / Cause	Occurrences	Total Hours
PCS	Inverter Alarm	514	25.4 h
BESS	LCU Alarm	15	290.1 h
BESS	BSC Unit Alarm (typo)	14	29.2 h
OTHER	Unclassified	13	13.4 h



Graph 5: Production alarm count and total downtime by raw device class. Used as a cross-check against the semantic classification above.

5.2 Major Incidents and Breakdowns

Top 10 longest production events (sorted by duration). Recurring multi-block cluster events indicate cascading faults and warrant joint RCA.

Activated	Element	Trigger	Duration
2026-04-02 20:34	System Info 13	LC - SYSTEM FAULT STATUS1 ACTIVE	46.6 h
2026-04-02 20:34	System Info 12	LC - SYSTEM FAULT STATUS1 ACTIVE	46.6 h
2026-04-02 20:55	BSC 13.02	BSC UNIT FAULT	46.3 h
2026-04-02 20:55	BSC 13.01	BSC UNIT FAULT	46.3 h
2026-04-02 20:55	System Info 13	LC-PCS COMM FAULT	46.3 h
2026-04-02 20:55	System Info 12	LC-PCS COMM FAULT	46.3 h
2026-04-02 20:55	System Info 13	LC-BSC FAULT	46.3 h
2026-04-02 20:34	System Info 11	LC - SYSTEM FAULT STATUS1 ACTIVE	46.2 h
2026-04-02 20:34	System Info 10	LC - SYSTEM FAULT STATUS1 ACTIVE	46.2 h
2026-04-02 20:55	BSC 12.02	BSC UNIT FAULT	45.8 h

6. Site Operation and Facility Management

Incidents affecting safety and countermeasures

Incident	Equipment Loss	Weight	Countermeasure
AC Breaker Fault	-	Low — no harm to personnel or major equipment damage	Reboot the PCS
Soft Start (DC switch) Fault	-	Low — no harm to personnel or major equipment damage	Reboot the PCS

Site visits and inspections

- 01.04.2026 — 13.04.2026: Sungrow service engineer and technicians visited the site to support the O&M; team during the planned auxiliary power shutdown. After restoration of aux supply they remained on site to monitor equipment performance.

7. Conclusions and Recommendations

1. Fleet round-trip efficiency was **92.27%** (above the 85% target). Container-level availability was **77.91%**.
2. Across 15 blocks, the fleet delivered **6,929.1 MWh** of discharge energy on **7,509.5 MWh** of charge, equivalent to **705 EFC** (avg 47.0 per block, 7.8% of the 600/year target).
3. **3,980** production-impacting events were recorded after deduplication (plus 9,342 persistent warnings, 24,460 transient events filtered as noise).
4. **3** block(s) flagged as underperforming versus fleet mean (RTE or throughput z-score < -1).
5. **1.9** day(s) per block flagged as EXCLUDED (zero throughput on both sides). These have been removed from RTE and availability calculations.

Recommendations & Mitigation Strategies

- Auxiliary power: consistent supply is a critical dependency — investigate alternatives to DG-only fallback for future outages.
- Shutdown procedure: reduce power setpoint to 0 before suspending charging/discharging. Power down only once PCS is in Standby.
- Start-up procedure: start all blocks first, confirm Standby, then issue power setpoints from SCADA PPC screen.
- Idle management: keep all blocks in Standby mode during non-operating intervals and transitions between charge/discharge cycles.

Planned Activities for Next Reporting Period

- PM activities scheduled for 12.05.2026 — 26.05.2026 (actual dates subject to site conditions).
- Continue daily performance monitoring and flag availability / RTE anomalies to Sungrow for rapid response.

8. List of Annexes

- Annex 1: Daily Block KPI table — full month per block (CSV export)
- Annex 2: Alarm classification lookup applied to this report (CSV)

- Annex 3: Quality-flagged days (EXCLUDED / INCOMPLETE / ESTIMATED)
- Annex 4: Top-block performance anomaly z-scores

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